Tabular Q Learning Lesson Plan

45 minutes

Preparation and materials

1. Slides 1–21
2. Board, paper, or iPad
3. charger_grid.py

Before class · Gymnasium (5 minutes). Gymnasium runs the environment and reports what happens after each move. In charger_grid.py, reset() starts the robot at A. Call step(action) to get the next cell, the reward, and whether the episode has ended. The Python Q-table uses that information to update the value of the move.

Instructor setup: install Gymnasium, run the two-episode demo, and have the slide 19 results ready to project.

Lesson stage	Time	Activities
Bridge In	0–2 min	Slide 1. Show the robot and charger. Ask how a robot that has just finished cleaning might learn to dock without being given a route.
Learning Outcomes	2–3 min	• Trace a grid move and identify its reward and next state. • Calculate a Q-update for a move that continues the episode and a move that ends it. • Complete the Python line that updates a Q-value and explain why exploration can uncover untried moves.
Pre Assessment	3–6 min	Slides 2–4. Ask pairs to find two routes from A to D without revisiting a cell. Invite students to list their routes.
Participatory Learning	6–18 min	Rewards, Discounting, and the Q-Table. Slides 5–9. Learners mark which move earns +10 and compare the direct route from B with the detour. Ask why both earn 10 before showing how $\gamma = 0.9$ makes their returns 10 and 8.1. Then have learners predict a wall move from A and help build the zero-filled Q-table.
Participatory Learning	18–28 min	Updating by Hand. Slides 10–15. Model episode 1, then have pairs calculate episode 2. Check for Q(A, Right) = 2.25 and Q(B, Down) = 7.5. If learners expect 0 after A → B, point them to B's best value at the time of the update.

Lesson stage	Time	Activities
Participatory Learning	28–32 min	Exploration. Slides 16–17. Show the route through C, then return to the Q-table after episode 2. Ask learners which move the robot would pick from A and what the zero for Down means. Introduce ε after they explain that Down is still untried.
Participatory Learning	32–37 min	Python Practice. Slides 18–19. Ask pairs to turn the Q-update formula from earlier into one Python line using old, ALPHA and target. Add their line to update() in charger_grid.py and run two episodes in Gymnasium to check it against the hand calculation. Then compare row C after 20 episodes at ε = 0 and ε = 0.5. Ask why Right from C is still zero.
Post Assessment	37–41 min	Slide 20. Individually, learners calculate the target and new Q-value for B → D. Give reward +10, old Q = 7.5 and α = 0.5. Look for target 10, updated value 8.75, and an explanation that D ends the episode. If someone adds a future value, ask what move the robot takes after docking.
Summary	41–42 min	Keep slide 20 up. Ask a learner to explain why the update into D uses only the immediate reward, then compare it with the earlier update from A to B.
Summary	42–45 min	Slide 21. Preview Deep Q-Networks (DQN) as a future topic. A 100 × 100 grid has 40,000 Q-table entries for four moves per cell. A DQN predicts four Q-values for the robot’s current state, one per move.